Activation Functions

Unit activation

$$\mathbb{R}^D \rightarrow \mathbb{R}$$

$$s(x|w, w_0) = w^T \cdot x + w_0$$

Sigmoid activation

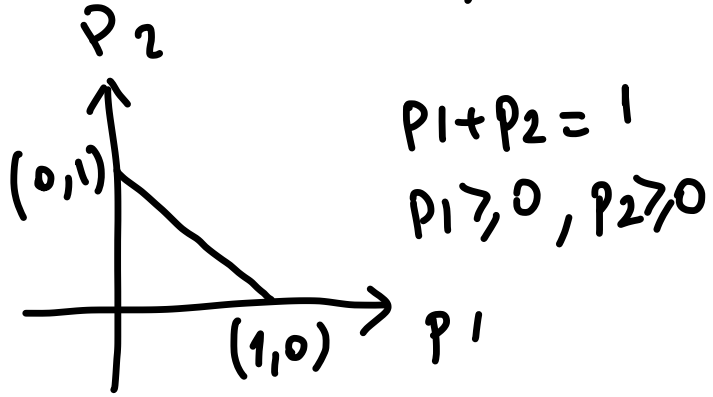
$$\mathbb{R}^D \rightarrow [0, 1]$$

$$s(x|w, w_0) = \frac{1}{1 + \exp\left[-\left[w^T \cdot x + w_0\right]\right]}$$

Softmax activation

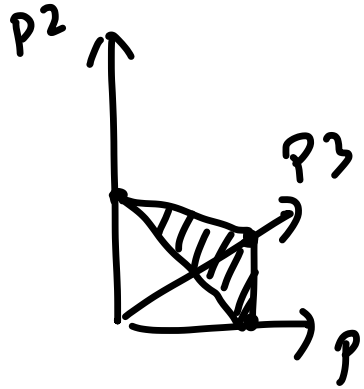
$$\mathbb{R}^D \rightarrow S^K$$

↳ simplex



$p_1 + p_2 = 1$

$p_1 \geq 0, p_2 \geq 0$



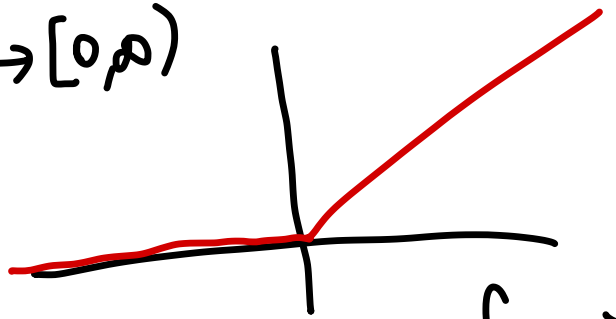
$p_1 + p_2 + p_3 = 1$

$p_1 \geq 0, p_2 \geq 0, p_3 \geq 0$

$$s(x \mid \{w_c, w_{c0}\}_{c=1}^{K}) = \begin{bmatrix} \frac{\exp(w_1^T x + w_{10})}{\sum_{d=1}^{K} \exp(w_d^T x + w_{d0})} \\ \vdots \\ \frac{\exp(w_K^T x + w_{K0})}{\sum_{d=1}^{K} \exp(w_d^T x + w_{d0})} \end{bmatrix}_{K \times 1}$$

ReLU (Rectified Linear Unit) activation

$\mathbb{R}^D \to [0, \infty)$



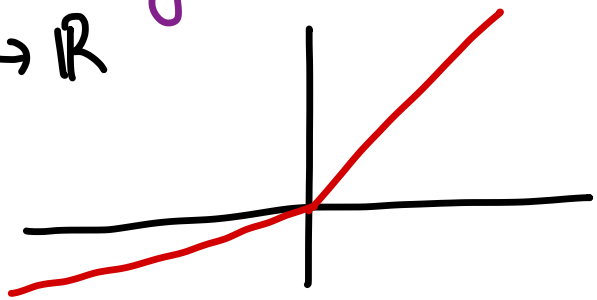
$$s(a) = \begin{cases} a & \text{if } a > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\underbrace{s(x \mid w, w_0)}_{\max(w^T x + w_0, 0)} = \begin{cases} w^T x + w_0 & \text{if } w^T x + w_0 > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$s'(a) = \begin{cases} 1 & \text{if } a > 0 \\ 0 & \text{otherwise} \end{cases}$$

Leaky ReLU activation

$\mathbb{R}^D \to \mathbb{R}$



$$s(a) = \begin{cases} a & \text{if } a > 0 \\ \alpha a & \text{otherwise} \end{cases}$$

↳ usually 0.001 or 0.01

$$s'(a) = \begin{cases} 1 & \text{if } a > 0 \\ \alpha & \text{otherwise} \end{cases}$$

Hyberbolic tangent activation (tanh)

$$\mathbb{R}^D \to [-1,+1] \quad \tanh(x \mid w, w_0) = \frac{\exp(w^T x + w_0) - \exp(-(w^T x + w_0))}{\exp(w^T x + w_0) + \exp(-(w^T x + w_0))}$$

$$\tanh(a) = \frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)} = 2\,\text{sigmoid}(2a) - 1$$

$$\tanh'(a) = [1 - \tanh(a)][1 + \tanh(a)] = 1 - (\tanh(a))^2$$